Silicon photomultiplier tube — Solution

Y23 (2023) — English translation

A1 **0.40 pts**

For steady state, find: voltage V_r at load r; voltage V_0 on each cell and parasitic capacitance C_0; voltage V_1 on the damping layers of each cell.

For steady state, find:

In steady state, no current flows through r, so $V_r = 0$. The voltage on the cells coincides with the source voltage, that is, $V_0 = V_{op}$. The capacitor C_1 is parallel to the resistor, so it is discharged: $V_1 = 0$.

Answer:

$$V_r = 0$$

$$V_0 = V_{op}$$

$$V_1 = 0$$

A2 **0.80 pts**

Find: change in charge Δq_0 of capacitor C_0. Express your answer using C_0 and ΔV; changes in charges Δq_{1a} and Δq_{2a} of capacitors C_1 and C_2, respectively, of each of the active cells. Express your answers using C_1, C_2, V_{ov} and ΔV; changes in charges Δq_{1p} and Δq_{2p} of capacitors C_1 and C_2, respectively, of each of the passive cells. Express your answers using C_{cell} and ΔV.

Find:

By definition of capacity: $\Delta q_0 = C_0 \Delta V$. The charge change on capacitors in active cells is similar: $\Delta q_{1a} = C_1(V_{ov} + \Delta V)$ and $\Delta q_{2a} = -C_2 V_{ov}$, and in passive cells: $\Delta q_{1p} = \Delta q_{2p} = C_{cell}\Delta V$.

Answer:

$$\Delta q_0 = C_0 \Delta V$$

$$\Delta q_{1a} = C_1(V_{ov} + \Delta V)$$

$$\Delta q_{2a} = -C_2 V_{ov}$$

$$\Delta q_{1p} = \Delta q_{2p} = C_{cell}\Delta V$$

A3 **0.70 pts**

Find the change in voltage ΔV across the cells and capacitor C_0 upon completion of the avalanche process. Express your answer in general terms through V_{ov}, n, N, C_1, C_2, C_0.

Let us write down the charge conservation law for positive plates C_0 and C_1.

$$n\Delta q_{1a} + \Delta q_0 + (N-n)\Delta q_{1p} = 0$$

$$nC_1(V_{ov} + \Delta V) + C_0\Delta V + (N-n)C_{cell}\Delta V = 0$$

$$\Delta V(nC_1 + C_0 + (N-n)C_{cell}) = -nC_1V_{ov}$$

$$\Delta V = -\frac{nC_1V_{ov}}{nC_1 + C_0 + (N-n)\dfrac{C_1C_2}{C_1+C_2}}$$

Answer:

$$\Delta V = -\frac{nC_1 V_{ov}}{nC_1 + C_0 + (N-n)\dfrac{C_1 C_2}{C_1 + C_2}}$$

A4 — 0.60 pts

Estimate the amplitude of thermal noise across the load $r = 50\ \Omega$ at room temperature if the oscilloscope bandwidth is $1\ GHz = 10^9\ Hz$. Using the data from the problem statement and the formula you obtained from A3, find the numerical value of ΔV when light hits one cell. Conclude whether the signal from one cell can be distinguished against the background of thermal noise of the load resistance.

Substitute $T = 300\text{K}$ and get: $V_{noise} = 2.88 \cdot 10^{-5}\text{V}$. Let us express the change in voltage through the data in terms of magnitude:

$$\Delta V = -\frac{(k+1)(1-p)\dfrac{n}{N}V_{ov}}{1 + \dfrac{kn}{N}(1-p)}$$

Substitute $n = 1$ and obtain:

$$\Delta V = -2.16 \cdot 10^{-4}\text{V}$$

Note, that ΔV in magnitude noticeably larger than V_{noise}, so the signal can be distinguished.

Answer:

$$V_{noise} = 2.88 \cdot 10^{-5}\text{V}$$

$$\Delta V = -2.16 \cdot 10^{-4}\text{V}$$

ΔV is noticeably larger in absolute value than V_{noise}, so the signal can be distinguished.

B1 — 0.80 pts

Find the multiplication coefficient M for the TFU with the parameters from the problem statement. Express your answer in terms of N, C_1, C_2, C_0, e, V_{ov} and give a numerical value.

Q can be determined through the change in charges on the negative plates C_0 and C_2.

$$Q = -C_0\Delta V - (N-1)C_{cell}\Delta V + C_2 V_{ov}$$

$$M = \left((C_0 + (N-1)C_{cell})\frac{C_1 V_{ov}}{C_1 + C_0 + (N-1)C_{cell}} + C_2 V_{ov} \right) \cdot \frac{1}{e}$$

$$M = 32.4 \cdot 10^3$$

Answer:

$$M = \left(\left(C_0 + (N-1)\frac{C_1 C_2}{C_1 + C_2} \right) \frac{C_1 V_{ov}}{C_1 + C_0 + (N-1)\dfrac{C_1 C_2}{C_1 + C_2}} + C_2 V_{ov} \right) \cdot \frac{1}{e}$$

$$M = 32.4 \cdot 10^3$$

C1 — 0.40 pts

Let's consider a pulse recorded by an oscilloscope after light hits one cell. We will assume that the pulse duration is short enough to neglect the current flowing through the leakage resistance R, but large enough so that the avalanche process is completed and all switches K_i are open. In this approximation, the circuit can be replaced by an equivalent series RLC circuit. Specify its parameters.

Answer: Resistance r, inductance L and capacitance C_{det}

C2 — 0.30 pts

Estimate the quality factor of the resulting RLC circuit. Will oscillations occur in this circuit?

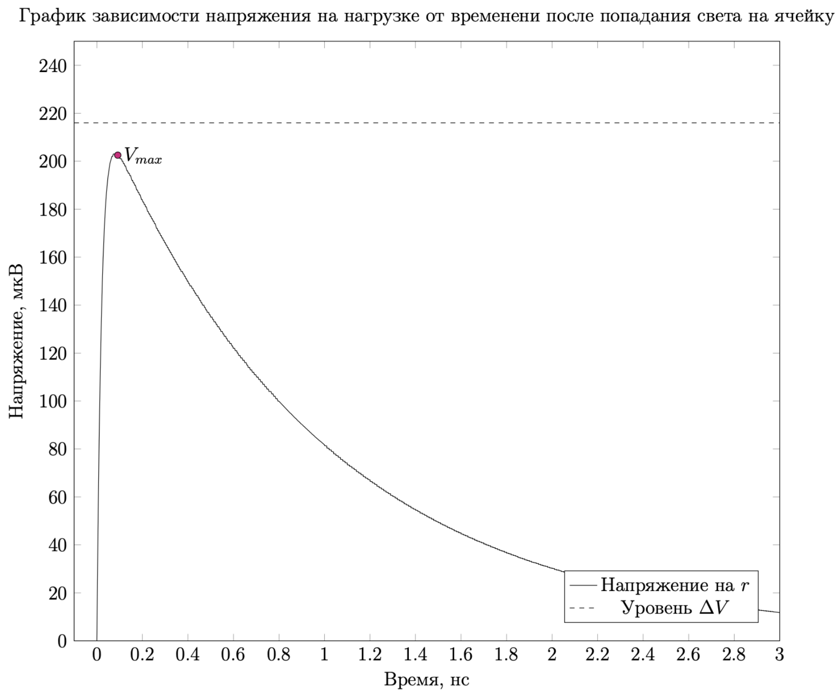
Answer: Let's use the usual formula for the quality factor of an oscillatory circuit:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = 0.14$$

The resulting value is much less than 1, which means that oscillations will not occur in this circuit.

C3 — 0.30 pts

Qualitatively depict a graph of the load voltage versus time after light hits one of the cells. Initially, the circuit was in steady state. It is not necessary to indicate characteristic values.



C4 — 0.20 pts

Compare the maximum absolute value of the voltage at the load $|V_{max}|$ after light hits one of the cells and the value $|\Delta V|$ found earlier. Which of these quantities is greater? Circle the correct ratio on your answer sheet.

The voltage across the resistor will always be less than the initial voltage of the capacitor.

Answer:

$$|V_{max}| < |\Delta V|$$

D1 — 1.00 pts

Find the approximate dependence of the voltage on the cell's damping layer on time after light hits it (under initial steady-state conditions). Take advantage of the fact that in the conditions of this problem $\Delta V \ll V_{ov}$, $R \gg r$. Consider that a lot of time has passed and the voltage on the external load is almost zero. Express your answer in terms of V_{ov}, R, C_1, C_2, t.

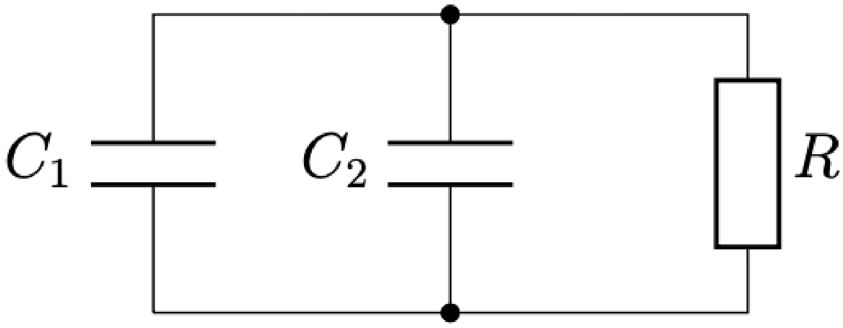
Due to the data relationships in the condition, the voltage at r and L can be neglected. That is, for the restoration process the cell is equivalent to an RC circuit.

Initially the voltage at C_1 is V_{ov}. Then time dependence:

$$V_1 = V_{ov} \cdot e^{-\dfrac{t}{R(C_1 + C_2)}}$$

Answer:

$$V_1 = V_{ov} \cdot e^{-\dfrac{t}{R(C_1 + C_2)}}$$



D2 — 0.50 pts

Find the change in voltage ΔV across capacitor C_0 when light hits one ($n = 1$) of the unrecovered cells, the voltage on the quenching layer of which was V_1.

The answer can be obtained in a similar way to part A, but now the change in voltage by C_2 will be $V_{ov} - V_1$ instead of just V_{ov}.

Answer:

$$\Delta V = -\frac{C_1(V_{ov} - V_1)}{C_1 + C_0 + (N-1)\dfrac{C_1 C_2}{C_1 + C_2}}$$

E1 — 0.30 pts

Let $p(t)$ be the probability that during time t the cell did not fire even once. Find $p(t + dt)$, express the answer in terms of $p(t)$, dt, T. Consider that $dt \ll T$.

The probability that the cell will not work in time dt is $1 - \dfrac{dt}{T}$, therefore:

$$p(t + dt) = p(t) \cdot \left(1 - \frac{dt}{T}\right)$$

Answer:

$$p(t + dt) = p(t) \cdot \left(1 - \frac{dt}{T}\right)$$

E2 — 0.20 pts

Find the probability that the cell will not fire even once during time t. Express your answer using t, T.

Let's use the result of the previous paragraph.

$$dp = -p \cdot \frac{dt}{T}$$

$$\frac{dp}{p} = -\frac{dt}{T}$$

Use $p(0) = 1$, we get:

Answer:

$$p(t) = e^{-\frac{t}{T}}$$

E3 0.20 pts

Let the cell fire at time $t = 0$. Find the probability that the next time it will work in the time interval from t to $t + dt$.

The desired expression is the product of the probability that the cell will not work in time t, and the probability of operation in dt.

Answer:

$$p(t, t + dt) = e^{-\frac{t}{T}} \cdot \frac{dt}{T}$$

E4 0.40 pts

Find the average amplitude of the photoresponse pulses μ. Consider that different cells produce impulses independently of each other.

From the definition of average:

$$\mu = \int\limits_0^{+\infty} V(t) \cdot p(t) \cdot \frac{dt}{T} = \int\limits_0^{+\infty} \left(1 - e^{-\frac{t}{\tau}}\right) \cdot e^{-\frac{t}{T}} \cdot \frac{dt}{T} = \int\limits_0^{1} \left(1 - x^{\frac{T}{\tau}}\right) dx$$

Where $x = e^{-\frac{t}{T}}$.

$$\mu = 1 - \frac{1}{\frac{T}{\tau} + 1} = \frac{T}{T + \tau}$$

Answer:

$$\mu = \frac{T}{T + \tau}$$

E5 0.60 pts

Find the standard deviation of the photoresponse pulse amplitude σ. Express your answer in terms of τ, T. Note: The standard deviation can be calculated using the formula $\sigma = \sqrt{\mu(V^2) - \mu^2(V)}$, where $\mu(V^2)$ is the average amplitude squared, $\mu^2(V)$ is the average amplitude squared.

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$\mu(V)$ was found in the previous point, similarly we will find $\mu(V^2)$:

$$\mu(V^2) = \int\limits_0^1 \left(1 - x^{\frac{T}{\tau}}\right)^2 dx = 1 - \frac{2}{\frac{T}{\tau} + 1} + \frac{1}{\frac{2T}{\tau} + 1}$$

$$\mu(V^2) - \mu(V)^2 = \frac{1}{\frac{2T}{\tau} + 1} - \frac{1}{\left(\frac{T}{\tau} + 1\right)^2} = \frac{\frac{T^2}{\tau^2}}{\left(\frac{T}{\tau} + 1\right)^2 \left(\frac{2T}{\tau} + 1\right)} = \left(\frac{T}{T+\tau}\right)^2 \cdot \frac{\tau}{2T+\tau}$$

$$\sigma = \frac{T}{T+\tau} \cdot \sqrt{\frac{\tau}{2T+\tau}}$$

Answer:

$$\sigma = \frac{T}{T+\tau} \cdot \sqrt{\frac{\tau}{2T+\tau}}$$

E6 0.30 pts

Find SNR amplitudes of photoresponse pulses under the influence of uniform background illumination in the following cases: $\tau = 0$; $\tau \gg T$; $\tau = T$.

Find SNR amplitudes of photoresponse pulses under the influence of uniform background illumination in the following cases:

$$SNR = \sqrt{\frac{2T+\tau}{\tau}}$$

Answer:

$$SNR(\tau = 0) = \infty$$
$$SNR(\tau \gg T) = 1$$
$$SNR(\tau = T) = \sqrt{3}$$